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Text Mining for Energy Materials

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Abstract: ***The scientific and technological progresses of the chemical and materials science disciplines lead to a significant amount of numerical and textual data stored in the published papers and databases, which can now be efficiently handled via the text mining process and employed for new chemical and materials discovery and machine learning-assisted analysis. In this manuscript, we review recent progresses of the text mining-assisted materials researches, with a special attention on the energy materials. The text mining process, facilitated by the development of machine learning methods, efficiently establishes hidden relationships and predicts new energy material candidates, including those for solar cells, thermoelectrics, batteries and catalysts. The word vectorization method (word embedding), such as word2vec, is especially efficient for the materials science researches. In addition, the text mining process assists the understanding of the energy and materials technologies and industries. Suggestions to better utilize the text mining techniques for the chemical and materials researches are provided. This manuscript is expected to facilitate the progress of the chemical and material researches via the text-mining method.***

Keywords: Text mining, Machine Learning, Artificial Intelligence, Energy generation, Informatics.

1. Introduction

The scientific and technological progresses of the energy materials lead to a significant amount of numerical and textual data stored in the published papers and databases, which can now be efficiently handled via the text mining process and employed for new knowledge establishment. The database selection is crucial for the materials screening process for a chemical, materials or device research. The natural language processing techniques offer viable ways to help extract information from the literature. As a result, it is important to take advantage of the natural language processing and text mining technique to obtain as much as accurate literature information as possible, followed by extracting the useful information and establishing hidden relationships between particular materials and research aims. The text mining method, which we will explain in detail, is suggested to unveil the hidden relationships behind the texts and predict the viable materials for certain applications effectively[1-10]. The relationships between materials and their functions are often hidden behind the texts and remain to be mined. When developing new materials, researchers often lack of sufficient understanding on the relationships between the materials and their application, property and performance, as well as the relationships between different technologies for trend analysis and technology forecast. We argue that the text mining method has capabilities to provide the platform to help the computers mimic the human cognitive process successfully by reading the research papers and learning the science and auto-predict the future in terms of the materials, technologies and industrial prediction. One of the great advantage is that the researches from a distant field can facilitate the knowledge given in a particular field by introducing novel ideas and concepts[11]. In addition, the text mining process can greatly accelerate the materials discovery process by using the 24h computers to replace the human labor and intuition[2].

The global energy crisis and severe environmental and atmospheric issues urgently call for the development of renewable energy materials to substitute the traditional fossil fuels. Various energy materials and devices have been devised, including those for solar cells, lithium batteries, thermoelectrics, hydrogen fuel and catalysis[12-20]. Investigating the energy materials and technologies is usually expensive and time consuming; the trial and error process in the traditional energy and material researches cannot satisfy the modern industrial and social requirements. With the coming era of the big data and artificial intelligence, the data-driven method acts as a new paradigm to effectively develop energy materials and technologies.

In this manuscript, we review the progress of the text mining methods that facilitates the extraction of information on the materials researches. To this end, the text mining procedure, popular natural language processing techniques, different text sources, preprocessing and the major processing step are described. The resources and software for the text mining are listed. The applications of the text mining method for energy materials, including thermoelectric materials, solar cell materials, battery materials, materials for hydrogen fuels and catalyst are explained. The text mining method is especially useful for the synthesis plan. In addition, the analysis of various technologies in the energy sector facilitated by the text mining is discussed. Suggestions to better utilize the text mining methods for materials and energy sciences are given.

2. General Text Mining Procedures

The text mining process generally includes document preparation, preprocessing, major text processing step (named energy recognition and relation extraction) and results generation (Figure 1). Text mining the literature for materials and energy systems can be either manual or automatic (the more automatic and autonomous process is preferred)[21,22], in which the unstructured data are compiled into structured data of materials[23]. It starts with the document retrieval, followed by the preprocessing involving checking the format and character and making necessary text treatment before the essential relationship study and modeling stage. In some cases, the database is enriched by experiments and first-principles calculations to facilitate the materials screening[24]. After the materials documents are downloaded and collected, the

preprocessing stage, often related to text cleaning, sentence splitting, tokenization, spell checker, pos tagging, lemmatization and stemming, is required to standardize and clean unstructured input words into structured ones (Figure 1). Compared with the pdf version, the HTML format using the markup language is more ready to be used. It is important to divide the document into various sections and extract information correspondingly, including the texts, figures and tables. In addition, the text should be divided into a series of processing unit, which is defined by the sentence; such sentence identification consists of the major part of the sentence boundary disambiguation. Tokenization, segmenting each sentence into punctuations, component words, expression sequences and number, is one of the most essential steps in the preprocessing stage. For example, Natural Language Toolkit is used for English word segmentation while Jieba is used for the Chinese word segmentation. Stop words are deleted to improve the efficiency, since the deletion of which does not affect the overall meaning, including prepositions and pronouns such as “a”, “the”, “to” and “their”[25]. A corpus should be constructed in this stage.

After that, a text analysis is performed to extract useful information based on diverse models and techniques (Figure 1). In the major text processing stage, the texts are analyzed (classified, categorized or clustered) and the relationships between the words are extracted. In this process, the pixel, path, shape, character, word, paragraphs and documents are turned into science. Some materials- and chemistry-aware natural language processing tools should be employed to identify and exploit the numerical and text data. The named entity recognition and entity relation extraction are critical in the process**.** The **named entity recognition** step identifies and classifies unstructured texts in to specific targets such as names, dates, temperatures, locations, quantities, costs, etc., which is a necessary step that leads to the construction of the relationships between the materials names and their structures, synthetic steps, properties and performance[26-29]. Diverse name recognition concepts have been proposed, with both semi-supervised learning methods and neural networks employed to assist the named entity recognition process with high precisions[30-32]. Many researches are dedicated to making targeted materials into graphs to present better human-computer interactions regarding the materials prediction[33]. The named entity recognition can be rule-based, dictionary-based, machine learning (ML)-based, or a hybridization of these methods. In addition, the identification of specific items expressed in different languages is important, since various countries and researchers/students using different languages are dedicated to the development of the materials science and technologies[25]. The named entity recognition is applied to extract important summary information from the materials science literature, including inorganic material mentions, sample descriptors, phase labels, material properties and applications, as well as any synthesis and characterization methods[21]. In the materials science field, the named entity recognition refers to the identification of chemical entities that make up the materials. On the other hand, the properties and detailed compositions are often not explicitly presented and should be revealed. Different types of machine learning methods, including support vector machine, maximum entropy model, hidden Markov model, conditional random field model and neural networks, are employed to help the named entity recognition, with decent F-score and accuracies during the evaluation stage. Among them, the deep neural network method is superior in terms of the generalization and less dependency on selected features, which is more often a prohibitive problem of other methods. **Entity relation extraction** is the major step of the text mining to reveal the hidden relationships and materials prediction. It involves the modeling of the texts and establishes the relationships between the entity pairs. Machine learning of often employed to aid the information extraction, which can be supervised, semi-supervised or unsupervised. Neural networks are employed to facilitate the relation extraction step[34]. The entity relationship extraction step is either following the entity identification step or performed simultaneously with the entity identification step[25]. Various models can be applied to extract the features, such as the Vector SpaceModel, Word2vec, Term Frequency, TermFrequency-Inverse Document Frequency, Mutual Information, Expected CrossEntropy, Gini, LatentSemantic Analysis, StatisticalLanguage Model, NeuralProbabilistic Language Model, Continuous Bag of Words (CBOW) and Skip-gram model. The **word embedding** (such as word2vec, Glove and fastText) is often utilized to help the language modeling and feature learning[35]. Essentially, it transforms the texts into vectors, which can be dealt with mathematical formula to identify the hidden relationships. However, problems remain such as the diverse meaning of a single word and various revised forms of the word embedding models are available. The vectorization is a versatile concept in materials development, for instance the unsupervised machine Atom2Vec that learns the properties of atoms and represents them in vectors[36]. After the relation extraction step, the evaluation is critical to understand the accuracy[25]. The evaluation indexes consist of Precision, Recall, F-score, etc[23].

A final database consisting of the relationships between named entities is constructed for further analysis[25].[25] Depending on the research purpose (for example: prediction of new materials for a particular applications; understanding the structure-property relationships; predicting the research trend; identifying the potential technology applications), the final **results** are generated and new knowledge are discovered, providing insights for scientific and industrial communities. This helps the decision making in the materials, chemical and energy sectors[25].

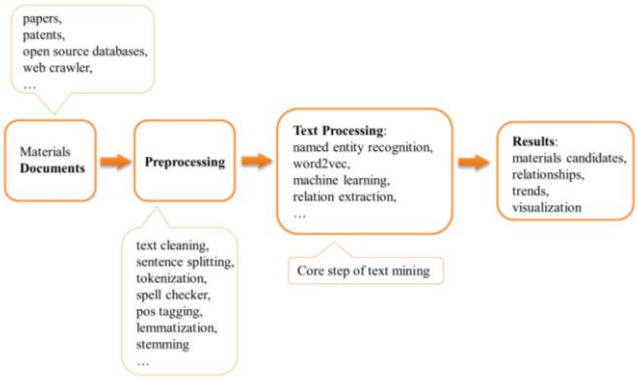

Figure 1: General procedure of text mining for materials science and technology.

The text mining, data mining, machine learning and natural language processing are mutually dependent. The text mining is essentially a data mining technique, with the data corresponding to the vast amount of texts in the literature. The data mining techniques are widely employed to automate and accelerate the materials discovery process[37,38,47-49,39-46]. In order to better handle the large amount of data to design materials and identify hidden relationships, **machine learning** is often employed[50,51,60-69,52,70-79,53,80-88,54-59]. The machine learning can either be supervised or unsupervised, depending on the extent of the human intervention. Quite often, neural networks are employed for the prediction purpose[89-95]. **Natural language processing** is the primary ingredient for the text mining process. It can be considered as a subdiscipline of artificial intelligent that assists the computers to efficiently understand, analyze and manipulate the human languages.

The processing of unstructured data into a structured form is of pivotal importance[96]. The starting databases for the text mining are generally unstructured and should be converted into structured data for machine-interpretation[97]. The manual collection and conversion of data is labor-intensive and comparatively less efficient[98]. Scientific data are most often stored in journal articles, industry reports and thesis. Among them, the **journal articles** represent the most scientifically valued formats. The published materials papers frequently used include those from Nature, AAAS, Wiley, ACS, RSC, Elsevier and SpingerLink[99]. The topics, authors, abstracts, keywords, references and entire contents have been considered for the text mining[27]. **Abstract** act as the predominant section to summarize the important findings in the articles and contains various materials names and chemical compounds and associated properties and performance[98]. Focusing on abstracts, information from millions of papers can be extracted and identify the materials relationship[97]. In contrast, the entire documents contain more information, but the more unstructured nature leads to a number of issues for the subsequent procedures. Published articles often contain information in the supporting information, which should be mined for further analysis. Citations are used to identify research fronts of various materials[100,101]. Bibiometrics involve using the statistical methods to analyze the data from publications, which is often encountered in the library and information science, such as those to predict the trend of the technology[102]. Application programming interface in various publishers should be utilized to tailor for the researcher requirements. In contrast, the **patents** focus more on the technology and industrial sides[103-106]. Text mining the patents is are important for the establishment of patent maps for topic analyses, patent classification, organization and knowledge sharing[107]. Apart from journal articles and patents, some general materials databases can be text mining for data analysis. For example, various structure databases stores information of crystal structures, including CSD for organic crystal and ICSD for inorganic crystal structures, in the format of “cif” files. Materials project stores calculated crystal information, materials names, tables of electronic and mechanical properties, and band gap values. Other databases that can be text mined include Wikipedia, lab books, thesis and industry reports[99]. Many structured ones can be found from the PubMed, Pubchem and SciFinder.

3. Resources and Software

Various tools and software for the text mining are available (Table 1)[108].

Table 1: Selected general and domain-specific tools and models for text mining

Name	Description	Website
Natural Language Toolkit (NLTK)	A Natural language processing tool developed by Bird and Loper from University of Pennsylvania	https://www.nltk.org/
StanfordNLP/Stanza	A Python-based natural language processing tool developed by Stanford NLP Group with more than 60 languages.	https://stanfordnlp.github.io/stanza/
Spacy	An open-source library based on Python and Cython for natural language processing	https://spacy.io/
AllenNLP	An open-source natural language processing tool	https://allennlp.org/
Gensim	An open-source library capable of unsupervised topic modeling and natural language processing, including fastText, word2vec and doc2vec algorithms	https://radimrehurek.com/gensim/
DataThief	Free software that extracts information from published work	http://datathief. org
ChemDataExtractor	A chemistry-aware text mining tool developed by University of Cambridge, automatically extracting chemical and material information from scientific literature.	http://chemdataextractor.org/
European Materials Modelling Ontology	A framework developed by the European Materials Modelling Council for a standard materials modelling and characterization ontology	https://emmc.info/emmo-info/
OSCAR4	An open-source chemistry analysis routines software with flexible platform for text mining	https://bitbucket.org/wwmm/oscar4/wiki/Home
ChemicalTagger	An open-source tool for tagging and parsing in chemical papers.	http://chemicaltagger.ch.cam.ac.uk/
ChemSpot	A chemical named entity recognition tool for trivial names and IUPAC entities	http://www.informatik.hu-berlin.de/wbi/resources
BioCreative	A joint effort for text mining chemical and biological information	https://biocreative.bioinformatics.udel.edu/
Chemlistem	A chemical named entity recognition tool via recurrent neural networks	https://pubmed.ncbi.nlm.nih.gov/30523437
ChemXSeer	A integrated library and database for searching and analyzing scientific articles	http://chemxseer.ist.psu.edu
CHEMDNER	A text mining tool for chemical name recognition	https://doi.org/10.1186/1758-2946-7-S1-S1
SciNER	A text mining tool for extracting named entities from scientific papers	https://doi.org/10.1007/978-3-030-50417-5_23
Word2vec	A word-embedding-based natural language processing algorithm using neural networks for unsupervised learning hidden information	arxiv.org/abs/1301.3781
Glove	Another unsupervised learning algorithm for word vectorization	https://nlp.stanford.edu/projects/glove
fastText	An open-source library for learning word embeddings developed by Facebook’s AI lab.	https://fasttext.cc/

The Natural Language Toolkit (NLTK), developed by Bird and Loper from University of Pennsylvania (https://www.nltk.org/), is a popular English-based natural language processing tool. In addition, Standford CoreNLP (https://nlp.stanford.edu/)[109], spaCy (https://spacy.io/) and AllenNLP (allennlp.org)[110] are also employed. Gensim is an open-source library for natural language processing, with fastText, word2vec and doc2vec algorithms. Other language-based processing tools exist such as Jieba for Chinese[23,111]. materials- and chemistry-aware toolkit include ChemDataExtractor[112], OSCAR4[28], ChemicalTagger[113], ChemSpot[32], Chemlistem[30], ChemXSeer[114], SciNER[26], CHEMDNER[31,115], European Materials Modelling Ontology (EMMO) and Materials Science Procedural Text Corpus, each having particular strength in chemical and materials name recognition and accessible text mining systems. These versatile materials-aware toolkits help preprocess and process the materials articles, and extracts names, structures, their properties, performance and synthetic procedures accurately using the natural language processing and machine learning methods[116]. The projects and resources for material synthesis include the Materials Science Procedural Text Corpus[117] and The Synthesis Project. Word2vec, developed by Tomas Mikolov in a Google team in 2013[118], relies on neural networks and vectorizes the words for easier mathematical treatments, and serves as one of the most effective toolkit to process the words in the literature[118]. Word2vec consists of two structures (skip-gram and cbow), depending on the relative positions of the starting words and the targeting words (skip-gram starts from the middle word and predict neighboring words; cbow starts from the neighboring words in the context and predict the middle word)[119]. Apart from word2vec, Glove and fastText are two other frequently used models for the natural language processing. The vectorization concept is applied to a number of physical researches, including those for the atomic structures[36]. The generation of a structured materials database is of critical importance. A large database of thermoelectric materials have been prepared, and visualization is provided to facilitate the screening[120]. Retrieval of both organic and inorganic materials for database construction has been achieved[98,121].

4. Examples of Text Mining for Energy Materials

4.1 Thermoelectrics

Tshitoyan et al. employ unsupervised learning techniques for word embeddings and successfully predict **thermoelectric materials** verified successfully after a certain year using only published papers before that year[97]. Such unsupervised learning of the literature information relies on the vectorization of the words in the articles, which are then trained to establish the relationships between the input materials words and the output application words via the cosine similarity. The word2vec-based method successfully replicates the periodic table and the structure–property relationships of materials without the human supervision, while various advanced materials and devices are clustered and grouped together automatically according to their types (Figure 2a). Interestingly, the experimental results obtained after the year 2005 successfully confirm the predicted thermoelectric materials using the literature published before that year (Figure 2b). This is first example of the unsupervised learning of the published papers using word2vec for the materials prediction purpose. Gaultois et al. carry out researches to aggregate and visualize the experimental data of thermoelectric materials[122], and prepare a database of thermoelectric materials consisting of 18,000 data points via text mining >100 articles (Figure 2c). Various information of the thermoelectric materials, including their the names, composition, crystal form, synthesizing method, publication details and power factors are obtained (http://www.mrl.ucsb.edu:8080/datamine/thermoelectrics.jsp)[120].

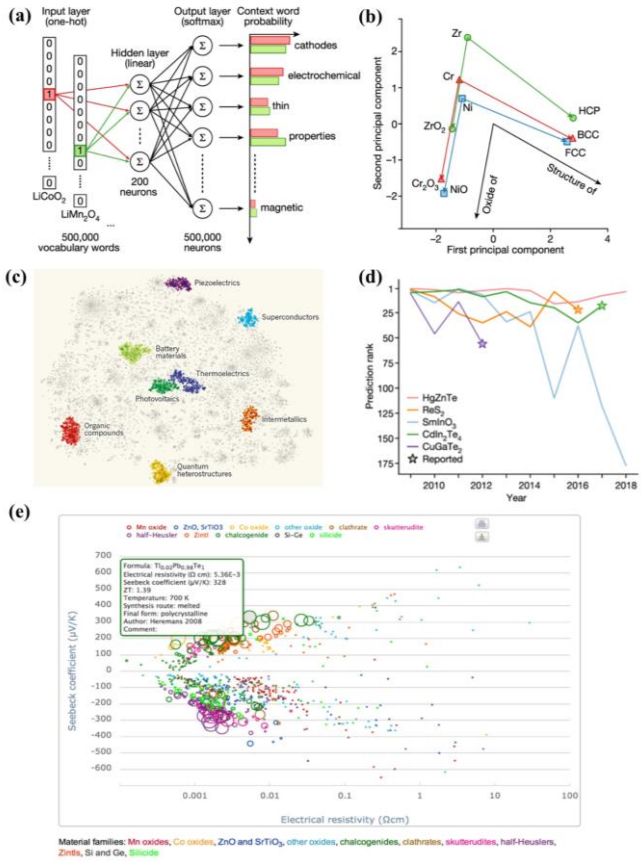

Figure 2: (a) Establishing the cosine similarity-based relationships between materials names (inputs) and their properties/performance (output) via word embeddings. (b) The validity of the text vectorization method exhibited by the parallel lines related to the "oxide of" and "structure of" various elements[97]. Reproduced with permission from Refs[97]. Copyright Springer Nature (2019). (c) Thermoelectric materials distribute together in the in the materials map obtained via the text vectorization method; the materials clustering also happens for superconductors, battery materials and organic compounds[2]. Reproduced with permission from Refs[2]. Copyright Springer Nature (2019). (d) Text mining the articles published before a particular year successfully predict the experimentally verified thermoelectric candidates in the future[97]. Reproduced with permission from Refs[97]. Copyright Springer Nature (2019). (e) Screenshot of a website containing thermoelectric materials and their properties after text mining the literature[120]. Reproduced with permission from Refs[120]. Copyright ACS (2013).

5.2 Solar Cells

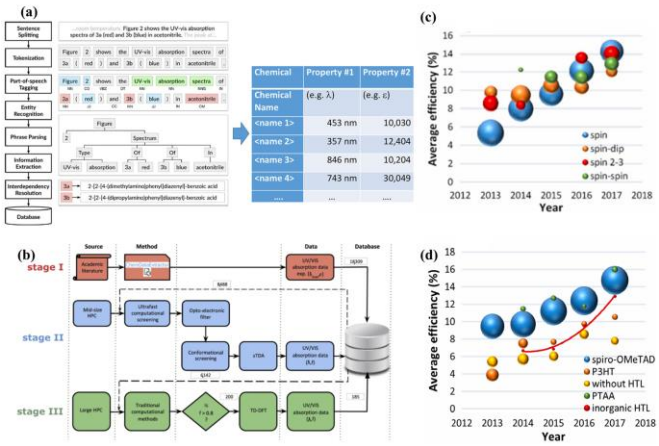

Figure 3: The natural language process to analyze texts for preparation of materials chemistry databases using ChemDataExtractor[24]. Reproduced with permission from Refs[24]. Copyright ACS (2020). (b) Stages to prepare comparative dataset of experimental and computational attributes of UV/vis absorption spectra. Reproduced with permission from Refs[123]. Copyright Springer Nature (2019). (c) Construction of perovskite solar cell database for analysis of the preparation methods[126]. Reproduced with permission from Refs[126]. Copyright Elsevier (2019). (d) Construction of perovskite solar cell database for comparison of common hole transporting layers[126]. Reproduced with permission from Refs[126]. Copyright Elsevier (2019).

The text mining process has been widely employed to extract **solar cell** information. The efficient UV-vis light absorption capability of the active layer, especially in the UV-vis region, is a prerequisite for the solar cell. The energy band gap (E_g) should neither be too large (transparent to UV-vis light) nor too small (energy loss and recombination) to satisfy the optical criterion. The UV-vis absorption information can be obtained via the natural language process tool ChemDataExtractor (Figure 3a)[24]. Beard and Cole et al[123]. prepare both experimental and computational datasets of UV/vis absorption spectra of 18,309 records from 402,034 scientific documents (Figure 3b), with the λ_{max} (peak wavelength) and f (oscillation strengths) from the literature compared with the simplified Tamm-Dancoff (sTDA) and time-dependent density functional theory (TDDFT) calculations[123]. For dye-sensitized solar cells, the text-mining method is employed to collect 9431 dye candidates with known optical absorption properties to offer desired dye combinations for co-sensitized solar cells, and six dye combinations are identified to present superior performance comparable to N719[124]. Such method is also employed to design the solar-powered windows because of the relevance of the dye-sensitized solar cells to the smart windows[125]. Alternatively, manually abstracting the perovskite solar cell information from the literature helps identify the detailed experimental ingredients. For example, 1921 **perovskite solar cell** data are obtained from 800 publications from 2013 to 2018 (Figure 3c), and analyzed via the machine-learning technique to understand the effects of the fabrication methods (solvents, anti-solvents, mixed cation utilization, spinning times, device architecture, etc.) and the materials selection on the electron transporting layer and the hole transporting layer on the overall efficiency (Figure 3d)[126]. In addition, 404 stability data from 181 published papers of perovskite solar cells are collected to understand their stability followed by the machine learning methods including association rule mining and decision trees.[127] However, the process is labor-intensive and could be improved via more automatic text mining methods. In addition, the text mining process is employed to connect the literature contents to the inorganic synthesis of perovskite compounds, via the successful prediction of their precursors[128]. The word2vec scheme is employed to predict the photovoltaic materials using historical datasets[97]. The top 50 solar cell material candidates predicted by the word embedding method demonstrate much higher probabilities to be experimentally validated as the actual solar cell materials compared with the materials without the word2vec-based text mining procedure. The materials predictions are based on the calculated cosine similarity with the target word "photovoltaics", which is highly indicated by the words such as solar, photovoltaic, PV, photodevices, photoelectronics, opto-electronics, LEDs, etc.

5.3 Lithium Batteries

A database consisting of **battery materials** is obtained via the text-mining 229,061 published papers, producing 17,354 materials and diverse properties including capacity, voltage, conductivity, Coulombic efficiency and energy (Figure 4a)[87]. Further analysis of the battery materials and their properties demonstrates the mutual correlations. Torayev et al. use text mining tools to review Li-O_2 batteries (Figure 4b) that enjoy remarkable capacities and demonstrate the progress of the battery dynamically[129]. Ghadbeigi et al. extract information from the literature manually using free software such as DataThief[130], where a large database of Li-ion battery material is prepared, indicating the consistency of the relationships between the cathode/anode materials and their crystal structures, as well as the relative scarcity of high energy cathodes and high performance anode materials (Figure 4c)[131]. The synthetic processes of the solid batteries are obtained from the literature, with the deep-learning method achieving a F1 score of 0.826[132].

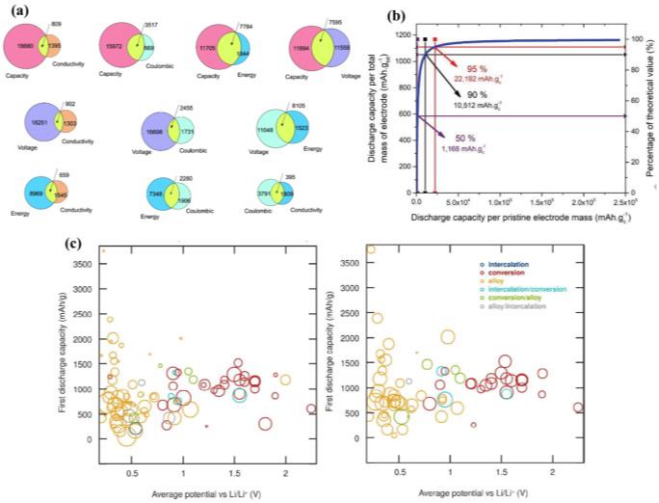

Figure 4: (a) Pair-wise permutation between the battery materials properties obtained via literature mining[87]. Reproduced with permission from Refs[87]. Copyright Springer Nature (2020). (b) Correlation between discharge capacities per pristine electrode mass and the discharge capacity per total mass of electrode[129]. Reproduced with permission from Refs[129]. Copyright IOP (2019). (c) First

discharge capacity versus the average potential, with the marker size proportional to the capacity retention after 25th cycle (left) and coulombic efficiency (right)[131]. Reproduced with permission from Refs[131]. Copyright RSC (2015).

5.4 Hydrogen Storage and Catalysts

Various metal-organic framework databases have been prepared for the text and data mining purpose for host and guest selection[131]. Tayfuroglu and Kocak et al. employ the text mining method to design the metal-organic frameworks for **hydrogen storage** (Figure 5a). The surface areas and pore volumes are comprehensively retrieved from the literature; these are compared with the structural calculations based on the crystal structures from the Cambridge Structure Databases, achieving a success rate of 78%-82% and efficiently predict the H_2 uptake values[134]. Various **catalysts** are designed to reduce CO_2, split water and produce solar fuels. Understanding the catalyst gene is critical for the material mining[135]. Ho and Hartman et al. employ the word embeddings to predict the catalysts and activator combinations for the metallocene polymerization[136]. Principal component analysis of word embeddings is performed for the metallocene catalysis polymerization (Figure 5b). The cosine similarity is then employed to understand the t-distributed stochastic neighbor embedding of 50 top-listed words to the words Cp2ZrMe2, and 1-hexene (Figure 5c)[136]. The text mining methods have been applied to a number of research areas, including chemistry[137,138], superconductors[98,139,140], energetic materials[141], quantum physics[142], robotics[143] and biomedicine[9,144,153,154,145-152].

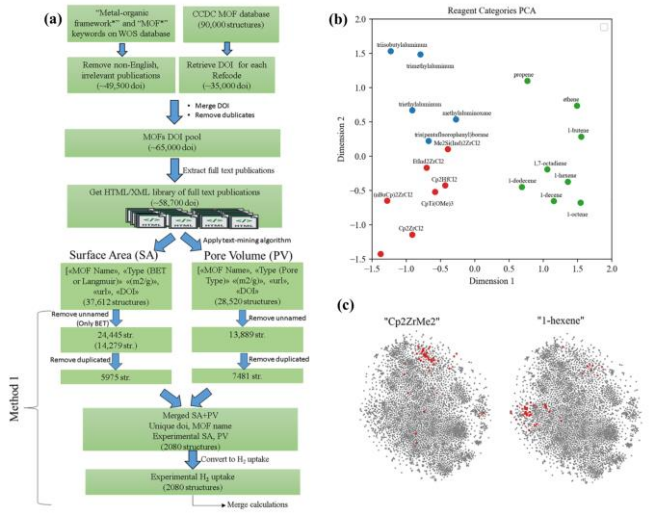

Figure 5: Text mining-based data-driven procedure to predict the H_2 uptake by metal-organic framework. The search starts from the papers from the Web of Science, which is compared with the crystal structures from the Cambridge Structure Database[134]. Reproduced with permission from Refs[134]. Copyright ACS (2020). (b) Principal component analysis of embeddings for: monomers, catalysts, and activators for the metallocene catalysis polymerization[136]. Reproduced with permission from Refs[136]. Copyright Elsevier (2020). (c) The cosine similarity is then employed to understand the t-distributed stochastic neighbor embedding of 50 top-listed words to the words Cp2ZrMe2 and 1-hexene. Reproduced with permission from Refs[136]. Copyright Elsevier (2020).

5. Examples of Text Mining for Materials Synthesis

The inclusion of the text mining process can guide the materials synthesis, simulating the way the human researchers are performing the literature review and planning for the energy materials synthesis. Hiszpanski and Han et al. employ machine learning-based text mining process to provide insights on the synthesis of nanomaterials, achieving prediction accuracies of 100% and 95% for composition and morphology, respectively, starting from the extraction from ~35,000 papers (Figure 6a)[155]. Synthetic procedures of inorganic materials are examined via the text mining process, with the dataset incorporating 19,488 synthetic procedure entries (Figure 6b) [156]. Kim et al. develop a machine learning-based text mining process to extract materials synthetic procedures to produce energy conversion and storage materials, including the synthetic procedures of metals oxides (titania) from 12, 000 papers (Figure 6c) [157]. Kim et al. prepare the synthesis database of oxide material using the grammatical treatment of a synthetic procedure sentence (Figure 6d)[158].

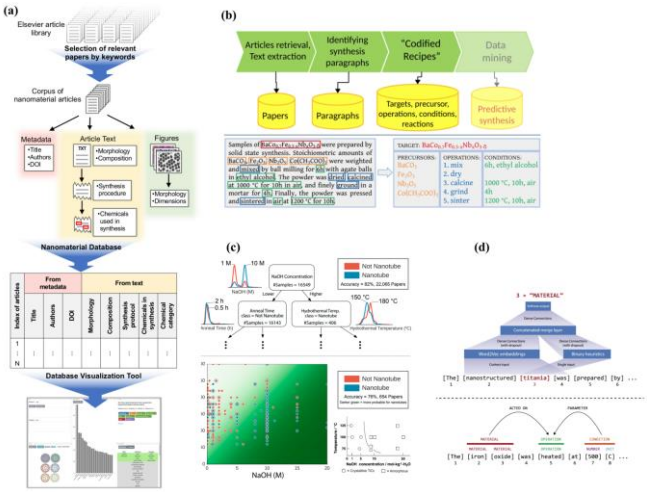

Figure 6: (a) Exemplars of the text mining studies for nanomaterials and related materials synthesis[155]. Reproduced with permission from Refs[155]. Copyright ACS (2020). (b) Pipeline to extract synthesis information, prepare synthesis database and predict the recipe designed by Kononova et al[156]. Reproduced with permission from Refs[156]. Copyright Springer Nature (2019). (c) Prediction of critical parameters in the synthetic procedures of titania based on text-learning method design by Kim et al[157]. Reproduced with permission from Refs[157]. Copyright ACS (2017). (d) Natural language processing technique to prepare the synthesis database of oxides starting from 640,000 papers designed by Kim et al[158]. Reproduced with permission from Refs[158]. Copyright Springer Nature (2017).

The deep learning method is employed to virtually screen inorganic materials synthesis, despite of the challenges regarding the data sparsity and data scarcity for certain materials; in this process the data augmentation including the synthetic data of related materials is crucial[159]. Vaucher et al. employ a deep-learning method to transform the unstructured synthetic steps to structured ones that are more accurate, and achieves a 100% sequence match for 60.5% of sentences[96]. Kim et al. train the literature with neural networks for

synthetic planning of inorganic materials, which is evidenced by the successful prediction of precursors for several perovskites using the papers published decades ago[128]. A semi-supervised machine-learning method based on a random forest algorithm is employed to classify the synthetic procedures of inorganic materials[160]. An automatic text mining method to facilitate the zeolite synthesis is proposed by Jensen et al., achieving minimal error based on zeolites containing germanium[161]. Attention is paid to the synthesis of oxide materials via pulsed laser deposition, starting from the related papers regarding the synthesis and property characterization of the materials (https://github.com/ ORNL-DataMatls)[162]. Understanding the reactant-based synthetic procedure structure is important for the design of materials[163,164]. A two-step chemical named entity recognition is employed to study the precursors of the solid-state synthesis, and the similarities of different precursors in terms of their impact on the final product can be quantified, such that the reactants can be better designed[165]. The materials fabrication steps should be annotated accurately to plan synthesis and understand the synthetic stages[117]. The merge of historical, experimental and computed data benefit the materials synthesis plan, and the varied model designed by Kim et al. is used to propose the synthesis of perovskite materials[128].

6. Examples of Text Mining for Energy Technologies and Industries

Text mining is widely employed to deeply understand the materials and energy research trend, technologies forecasting, industries and policies[105,166-170]. To this end, the patent data is extensively employed for the discovery and prediction purpose[171]. The neural embedding accelerates the organizations and classification procedure[172].

Promising **solar cell** research frontiers are explored based on the accumulated documents of papers and patents[173,174]. The text mining technique with the cluster analysis is employed to propose solar cell technology strategic planning, relating the companies and the techniques[175]. The library (database) is critical to evaluate the photovoltaic performance; for example, the addition of a particular MO_3 layer to the $TiO_2|Co_3O_4$ library affect the power conversion efficiency profile[176]. In order to bridge the gap between science and technology of solar cells, the citation networks of papers and patents are compared, confirming the different focuses of the two media. Only silicon and compound solar cells exhibit sematic similarities between the patents and papers, suggesting the immediate commercialization possibilities of the two systems[177]. The combination of text mining and bibliometrics are employed to analyze the **dye-sensitized solar cell** and organic solar cell industry[178-180]. Text mining and visualization of the dye-sensitized solar cell and thin-film solar cells are performed to analyze the research activity, and predict the future application[181]. The technical phrases of dye-sensitized solar cells are reduced to more user-friendly version, which is complemented and evaluated by knowledgeable persons in the field[182]. The forecast of technologies is assessed by revisiting previous analysis on dye-sensitized solar cells to understand the prediction accuracy[183]. Different text mining software are compared for this renewable energy technology[108]. Based on patent data, the hotspot technologies of **organic solar cells** are identified, involving the device architecture, morphology, electrode and charge transport layer, active layer, interface, assembly and packaging[184]. The evolution of hot topic of **perovskite solar cells** and organic solar cells from 2008 to 2017 is analyzed, indicating the importance of the fullerene-free and stability/degradation concepts[185]. The trend of the perovskite solar cell technology is proposed by patent and twitter data analysis, demonstrating the importance of the social media for the technology analysis[186]. The research trend of perovskite solar cells is identified by analyzing patents and web news[187].

Text mining is performed to review the lithium-ion batteries and lithium-oxygen **batteries**, demonstrating the trend of the development of the particular topic[122,129]. Technology development trend of lithium-ion battery derivatives such as the sodium-ion battery is analyzed by accumulating the patent information[188]. Material flow is analyzed for lithium battery materials[189] Text mining is performed to understand many other energy materials and technologies such as electrochemical power[190] solid oxide fuel cells[191] and nanogenerator technology[192].

7. Suggestions and Outlook

Since different types of materials databases have large impacts on the final result generation and the discrepancies between computationally and experimentally determined data values exist, it is important to compare and employ different types of databases including the simulation and experimental literature in the text mining researches. Combining multiple databases with various data types, sizes, languages, research areas and publishers is admired during the text mining process. In the meantime, less accurate databases and those offering weaker scientific insights should be avoided. It is well-accepted that the online websites such as Wikipedia predicts less accurately the proper materials candidates compared with the published papers. Unsuitable metadata types such as erratum and memorial should be excluded in the text mining process. Moreover, the word embedding algorithms such as word2vec treat the negation ineffectively, since the inclusion of NOT before certain words leads to totally different meaning. More rigorous natural language processing models should be adopted to better predict the materials and energy systems and unveil hidden relationships.

One promising direction of the text mining-driven scientific research for the materials and energy departments is the proper handling of the huge amount of information written in different languages from various countries. Although most of the valuable information in the literature is published in English, there are many resources providing relevant information written in another language, such as German, French, Chinese, Japanese, etc., used for dissertations and industrial reports (for example, Tesionline, Dissonline, Thèses et écrits académiques, Système universitaire de documentation, Karlsruher virtueller katalog, China doctoral/master dissertation database (CNKI), Vidyanidhi, Norsk bokfortegnelse and Bases de datos de tesis doctorales, etc.). In addition, the patents in a non-English version are especially valuable and have more impacts on the industries; these patents provided by diverse countries in different

languages should be mined efficiently to bridge the gaps among science, technologies and industries. The language barrier makes most researchers impossible to fully exploit the literature information; as a result, the text mining tools capable of dealing with non-English articles friendly for most scientists and engineers with minimal foreign language skills are required. These text mining tools should be correctly utilized, and a better "big-data" process is expected when the hidden information containing multiple languages can be extracted.

More efforts should be made to the exploit of the pictures (image mining) in the published articles, since "a picture is worth a thousand words", and he picture-based mining process complements the text mining methods.[193] When researchers are reading scientific articles before conducting real experiments, it is the pictures that come to their mind intuitively and are should be considered to be of primary importance. Eventually, we hope to use computers to substitute humans and automatically read the literatures (both texts and images) and conduct screening and relationship extraction processes for materials, chemical and energy researches. Several image mining and visualization software are available[193-198]. The acceleration of the image mining process depends on the image recognition techniques, not only involving the geometrical attributes but also the materials attributes[199,200].

Full automation of the text mining process with minimal human intervention is an honorable goal in the long run for these researches. Since various simulation and quantum mechanical software are available to calculate the materials properties, a "master code" that can execute the text mining procedure autonomously and perform the computer-based evaluation of the screened candidates is viable in the future. In addition, artificially intelligent robots that can text mine the literature itself and subsequently conduct experiments independently are desirable[201]. Recently, Mehr and Cronin et al. successfully demonstrate an architecture to read the literature automatically and automates the syntheses of twelve compounds from the literature[202]. This demonstrates the viability and necessity of integrating the text mining technique into the real-scenario experimental processes.

An unavoidable issue of the text mining process for every scientific research is the lack of free contents, which necessitates the consideration of legal issues and open access. To date, most high impact journals and patents require subscription fee. Meanwhile, efforts have been made to lower the literature access barriers by some scientific publishers[203]. The open access (OA) journals can be utilized to extract useful information to avoid the legal issues, at the expense of the author funding. In addition, the free content, especially the abstracts of a paper containing the essential ideas, materials and applications of the research, should be focused; it is advised that the authors provide rich and succinct information in the abstract (especially the detailed chemical and materials names) when they are submitting their manuscript in the future.

8. Conclusion

Exploiting the literature information via text mining is showing promise to identify new energy material candidates and reveal the hidden relationships between materials and their applications. In addition, the text mining method is effective to facilitate the understanding of the energy research trend and identify understudied area, suggesting emerging energy technologies and helping improving energy policies. The process mimics the literature reading and brainstorming process of human when conducting a research, and enjoys the semi-autonomous and automatic design process via computers. Further automation in the text mining process with minimal human intervention is required. Advanced robots bearing the text mining capabilities to read papers and patents, extract relationships and conduct simulations or experiments with improved autonomy and automation are called for. The combination of text and image mining, the utilization of diverse languages and the provision of more open sources can further advances in the text mining-based energy material and technology researches. The employment of the text mining method is expected to greatly facilitate the material, chemical and energy researches and industries.

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Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study

Conflict of Interest Statement

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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